

What the excitation data say about the bias

The coefficients above are predictions. The excitation runs test them without any quantity being fitted, through the gap that the rigid balance leaves.

The balance with the ground-effect channels restored At the onset the far contact unloads, and the moments about the loaded contact balance. Ground effect acts on the two terms the rotors produce and on neither of the others: the collective thrust is augmented near the ground, and the commanded moment is delivered with a gain. Gravity is not,

$$\underbrace{(1 + \eta_M^{GE}) M_{\text{cmd}}}_{\text{commanded}} + \underbrace{(1 + \eta_f^{GE}) f l_p}_{\text{thrust}} = \underbrace{W l_p + S_{\text{off}}}_{\text{gravity}}, \quad (1)$$

with l_p the contact arm and S_{off} the offset moment. Solving for the command gives the predicted threshold,

$$M_{\text{cmd}} = \frac{\text{sgn} [W - (1 + \eta_f^{GE}) f] l_p + S_{\text{off}}}{1 + \eta_M^{GE}}, \quad (2)$$

which reduces to (eq:mcrit`roll) and (eq:mcrit`pitch) when both channels vanish. Note the order in (eq:gepred): the thrust is corrected *before* the tipping requirement is formed, because what remains on the contact is $W - (1 + \eta_f^{GE}) f$ and not $W - f$.

The gap, and which channel supplies it Writing the no-ground-effect threshold as T_{rigid} and the detected onset moment as M_{det} ,

$$\Delta M \triangleq T_{\text{rigid}} - M_{\text{det}} \quad (3)$$

is the moment the ground supplies but the rigid balance omits. Over the twenty direction groups of the five configurations the vehicle tips at a *smaller* commanded moment than the rigid balance requires in every one of them, by 204 mN m in the median — about a fifth of the threshold itself.

The interference model of (eq:ge`moment) accounts for 165 of that, or 81%, with no tuned constant. That figure should be split, because it bundles two claims of very different standing. Linearising (eq:gepred), the thrust channel contributes $\eta_f^{GE} f l_p$ and the moment channel $\eta_M^{GE} M$:

	median	mean	share of ΔM
gap ΔM	204	202	—
thrust channel $\eta_f^{GE} f l_p$	117	115	57%
moment channel $\eta_M^{GE} M$	43	47	21%
residual	49	40	22%

in mN m. Nearly three-quarters of the modelled effect is the thrust augmentation acting through the contact arm — a measured force scaled by an established augmentation ratio — and only 21% of the gap is a ground-*induced moment*, which is the part that rests on the rotor-interference model. The speculative claim is therefore a fifth of the gap, not four-fifths of it.

One qualification belongs with the number. ΔM is not a measurement of the aerodynamic moment: it is what the rigid balance leaves over, and two of that balance’s ingredients are themselves modelled — the collective thrust is $\sum C_T \omega_i^2$ from a bench identification, hence an out-of-ground-effect value by construction, and the circle fit of Sec. V locates the *kinematic* pivot, whereas the balance needs the line of action of the normal-force resultant. ΔM therefore collects the ground’s total contribution, aerodynamic and contact alike.

Two checks that use no external reference Both (eq:weight) and (eq:offset) close on themselves. Eq. (eq:offset) estimates the same offset twice, once from each tip direction with that direction’s own arm, and (eq:weight) returns the weight from the moments. Neither closes without a ground-effect term: the two directional offset estimates disagree by 12.6 mm in the median (19.8 at worst), and (eq:weight) returns a weight 5.2% below mg , low in all ten case/axis configurations. Restoring the channels brings the directional spread to 3.6 mm (9.0 at worst) and the weight to 1.1% below; the thrust channel does most of the latter, since it enters (eq:weight) as $(1 + \eta_f^{GE}) f_{\text{crit}}$. Neither the load cell nor the identified offsets enter these two checks.

The gap is also not an artifact of the onset detector. The four model-agnostic baselines of Sec. II-D bracket the identified moment within a median 108 mN m, and all detectors place the per-run onset moment within 3–45 mN m of one another, both well below ΔM .

Table 1: Attribution of the threshold deficit over the twenty direction groups. H1 free lets the data choose the channel gains; H1 fixed uses the interference coefficients with a single scale.

hypothesis	fitted	residual RMS [mN m]
H1 free	$\eta_f = 0.098 \pm 0.027, \eta_M = -0.055 \pm 0.063$	78.7
H1 fixed	scale = 1.213 ± 0.12	84.8
H2	$\delta l = 19.4 \pm 1.9$ mm	83.6
H1+H2	const = 0.235 ± 0.06 N m, $\eta_M = -0.031 \pm 0.052$	76.0

The residual cannot be attributed, and three tests say why

The 49 mN m the model leaves is not a small coefficient error. Two physically distinct explanations fit it equally well, and the data cannot separate them.

H1, ground effect: the residual is a mis-sized version of the same two channels. *H2, contact geometry:* the normal-force resultant acts a distance δl inboard of the kinematic pivot circle, as a finite compliant contact patch implies, so the static lever is $l_p - \delta l$. Both scale as force $\times$ length and are degenerate in the constant part; the only discriminating structure is the moment-proportional channel of H1, which grows with $|M_{\text{crit}}|$ over a $4.3\times$ range across groups while H2 does not. Table 1 fits both.

Three things follow. Letting the gains free barely improves the fit, 84.8 to 78.7 mN m, so the form is not the problem. The discriminating gain is not detected at all: η_M comes back negative with an interval that spans zero, and the reason is visible in the scale — the $\eta_M M$ signal spans 18 to 79 mN m across groups while the group-to-group scatter of the deficit is itself 79 mN m. And H2 alone fits as well as H1 does, 83.6 against 84.8, with a lever displacement of 19.4 ± 1.9 mm.

That degeneracy is a property of the design, not an accident of the sample, and three independent attempts to break it with the data in hand all return null.

The rotation-centre height. If a compliant leg moved the instantaneous centre up from the foot, both the destabilising coefficient and the pivot inertia would shrink together, which is what the identified constants suggest. Freeing the pivot height in the circle fit and sweeping it leaves the arc residual unchanged to two decimals — 0.14 mm at the minimum against 0.14 at ground level — because a 10° arc trades the height against the radius almost freely. The arm itself does bound it, however: a centre 25 mm above the ground would return $l_p = 154$ mm against the 140.4 ± 3.6 measured, so the centre is at the foot within about 10 mm and this reading is excluded rather than merely unresolved.

Ballast dependence. Moving the ballast redistributes the load among the contacts, so if the residual were contact compliance its effective stiffness should track the offset. No component of the offset predicts it: the strongest correlation over the ten configurations is $|r| = 0.40$ at a permutation p of 0.25.

Contact migration during the arc. The threshold is defined at the onset, $\varphi = 0$, while the circle fit averages the arc that follows, so a contact that migrates outboard as the vehicle tips would inflate the fitted arm and over-predict the threshold — the observed sign. The migration leaves a trace, since the marker then traces a trochoid rather than a circle, and the observed fit residuals of 0.1 to 0.2 mm correspond to an arm bias of 9.6 to 18.4 mm, which is the right size against the 19.4 mm fitted. But it does not vary the right way: the correlation between residual and deficit is $+0.58$ pooled at $p = 0.081$, and it decomposes into $+0.75$ for roll against -0.67 for pitch, so the pooled figure is the axis offset rather than a relation.

A clearance sweep would separate H1 from H2, since the aerodynamic term scales with h/R and the contact term does not: repeating one axis at $h = 0.60$ m would put the aerodynamic hypothesis at 0.069 N m against the contact hypothesis's unchanged 0.20, a three-fold split well above the 0.06 N m floor. That sweep is outside the present dataset, and Sec. IX returns to it.

Why the deliverable does not need the attribution

The reviewer's concern is whether this bias corrupts the delivered offset. It does not, and the reason is structural rather than fortunate.

The lever error cancels by direction Writing the two directional thresholds with their own arms and taking the pivot-free average of (eq:pivotfree),

$$M_{\text{ff}} = \frac{1}{2}(M_+ + M_-) = \frac{1}{2}(W - f)(l_+ - l_-) + S_{\text{off}}, \quad (4)$$

Table 2: Identified CoM offset against the load-cell truth over the ten case/axis configurations, with and without the ground-effect channels of (eq:offge). “ Δ ” is the mean paired change in $|\text{error}|$ against the $\eta^{GE} = 0$ row of the same block, with the exact sign test beside it.

arms	ground effect	RMS	mean	max	Δ	p
		[mm]	[mm]	[mm]	[mm]	
pivot-free	none (reported)	1.64	−0.91	3.34	—	—
pivot-free	single-rotor	1.67	−0.98	3.44	−0.00	1.00
pivot-free	interference	1.82	−1.20	3.77	+0.19	0.11
pivot-based	none	1.77	+0.42	2.96	—	—
pivot-based	single-rotor	1.73	+0.35	2.80	−0.01	1.00
pivot-based	interference	1.65	+0.13	3.05	−0.06	1.00

so any error in the contact lever that is *common* to the two tip directions leaves M_{ff} untouched. The 19.4 mm of H2 is such an error, and it contributes nothing. Only the direction asymmetry survives, and it is small: decomposing the residual of Table 1 by tip direction gives an antisymmetric part of 54 mNm in the median, which cancels, against a symmetric part of 37, which does not. The latter corresponds to 7.0 mm of arm asymmetry through $dM/dl = W - f = 10.6$ N, hence 1.17 mm of offset — against the 1.43 mm between-configuration scatter of the offset error measured in Sec. VIII-C.

That agreement is the point. What the static check fails to attribute and what limits the delivered accuracy are the same quantity, seen from two sides, and it is set by the contact geometry rather than by the aerodynamics or by the estimator.

The geometry itself checks out The skid measures 286.29 mm across and 250 mm fore-aft, so the half-widths are 143.14 and 125.00 mm against circle fits of 140.5 ± 3.6 and 112.7 ± 6.6 . Roll agrees to 2.7 mm; pitch sits 12.3 mm inboard of the CAD edge, as a rail whose ends curve up should. Substituting the CAD half-widths makes the deficit worse — pitch 69 to 173 mNm — so the nominal arm is not the error, and the per-run fitted arm is the right one to use. A rigid rolling foot biases the fit in the right direction but by 0.4 to 0.9 mm for realistic radii, an order of magnitude short of δl .

Worth recording separately is that the measured arm asymmetry $l_+ - l_-$ is systematic per axis rather than scatter: negative in all five roll groups and positive in all five pitch groups, 7.6 mm in the median. The pivot-free form of (eq:pivotfree) discards it by assuming $l_+ = l_-$; the pivot-based form of (eq:offset) keeps it, which is why the two are reported side by side below.

Effect on the identified offset Restoring both channels in the onset balance and pairing the directions,

$$S_{\text{off}} = \frac{1+\eta_M^{GE}}{2}(M_+ + M_-) + \frac{\eta_f^{GE}}{2}(f_+l_+ - f_-l_-) - \frac{1}{2}[(W - f_+)l_+ - (W - f_-)l_-], \quad (5)$$

the last bracket being dropped in the pivot-free variant. Table 2 evaluates (eq:offge) against the load-cell truth at both levels of physics and both treatments of the arms.

The ground-effect term does not improve the offset, and no paired comparison reaches significance: the correction is 0.41 mm in the median against a 0.45 mm standard error of the comparison itself, so this dataset bounds the ground-effect contribution to the offset at about half a millimetre either way rather than resolving it. What does move systematically is the mean bias, which the fullest inversion — arms and interference together — removes, from −0.91 to +0.13 mm at unchanged RMS.

What is applied, and what is reported The two uses of the same measurement need separating. The compensation applied at take-off is M_{ff} of (eq:pivotfree) with no ground-effect model anywhere: it is the in-situ balanced moment and already carries whatever the ground contributes, which is precisely what has to be cancelled, so subtracting a modelled term there would remove part of the disturbance the compensation exists to reject. Reading the same quantity as a mass property is the separate question of Table 2, and the offset is reported at $\eta^{GE} = 0$ to match the quantity flown, with the table standing as the bound on what the omission costs.

The interval the excitation measures directly supports this. Any moment strictly inside $[M_-, M_+]$ keeps both contacts loaded, the midpoint maximises the smaller of the two margins, and both endpoints come from the excitation rather than from a model. So M_{ff} is the max–min–margin choice whatever the unmodelled terms turn out to be, and by (eq:ffcancel) the direction-common part of them — 54 mNm against a 1000 mNm half-interval — never enters it at all.